EXAMPLE UNIVERSITY - MID-SEMESTER EXAMINATION 2025
School of Engineering (synthetic sample paper)
Subject: Sample Systems Modelling (XX3001) (Regular)
> 4 (four) questions are to be attempted.
> Section A is compulsory; attempt any 1 question from each of Sections B, C and D.
Time: 1.5 hours Full Marks: 20
The figures in the right-hand side indicate full marks.

Compare a living nerve cell with a model unit.

How can the behaviour of a model unit be written down mathematically?

What does the winner-takes-all learning rule do?

Which kind of learning does vector coding by prototypes use?

Draw a fuzzy set to describe the statement the room is warm, over a continuous temperature axis.

Two sets are given as P = ((x1, 0.1), (x2, 0.2), (x3, 0.7), (x4, 0.5), (x5, 0.4)) and Q = ((x1, 0.9), (x2, 0.6), (x3, 0.3), (x4, 0.2), (x5, 0.8)). Determine the following. i) $P_{0.2}$ and $Q_{0.3}$ ii) The support of each set iii) The core of set P iv) Convexity of set Q v) Normality of set P

Take the sets L = {(low,1),(medium,0.2),(high,0.5)} and K = {(positive,0.9),(zero,0.4),(negative,0.9)}. Form the product relation R = L x K and then the diagonal of $R\circ R^{T}$.

A single linear unit cannot split classes that are not linearly separable. Justify this.

Train the OR function (binary inputs and targets) on one unit with one neuron for one epoch, with initial weights and bias w1=0.1, w2=-0.5, b=0.1 and learning rate 0.5.

Train the AND function (bipolar inputs and targets) on a two-input one-output adaptive unit from w1=0.2, w2=0.2, b=0.2 for one epoch with learning rate 0.5.

Design a radial basis network for a classification task and show that the problem becomes linearly separable in the hidden space, taking the XOR function as the example.

The output layer of a map is a ring of six nodes A to F (dashed links A-B-C-D-E-F-A). Each node has two inputs x1 and x2 (not shown). The initial weights are

a) Find the winning node for the input x = [2, -4].
b) Adjust the winning node's weights with a learning rate of 0.5.

